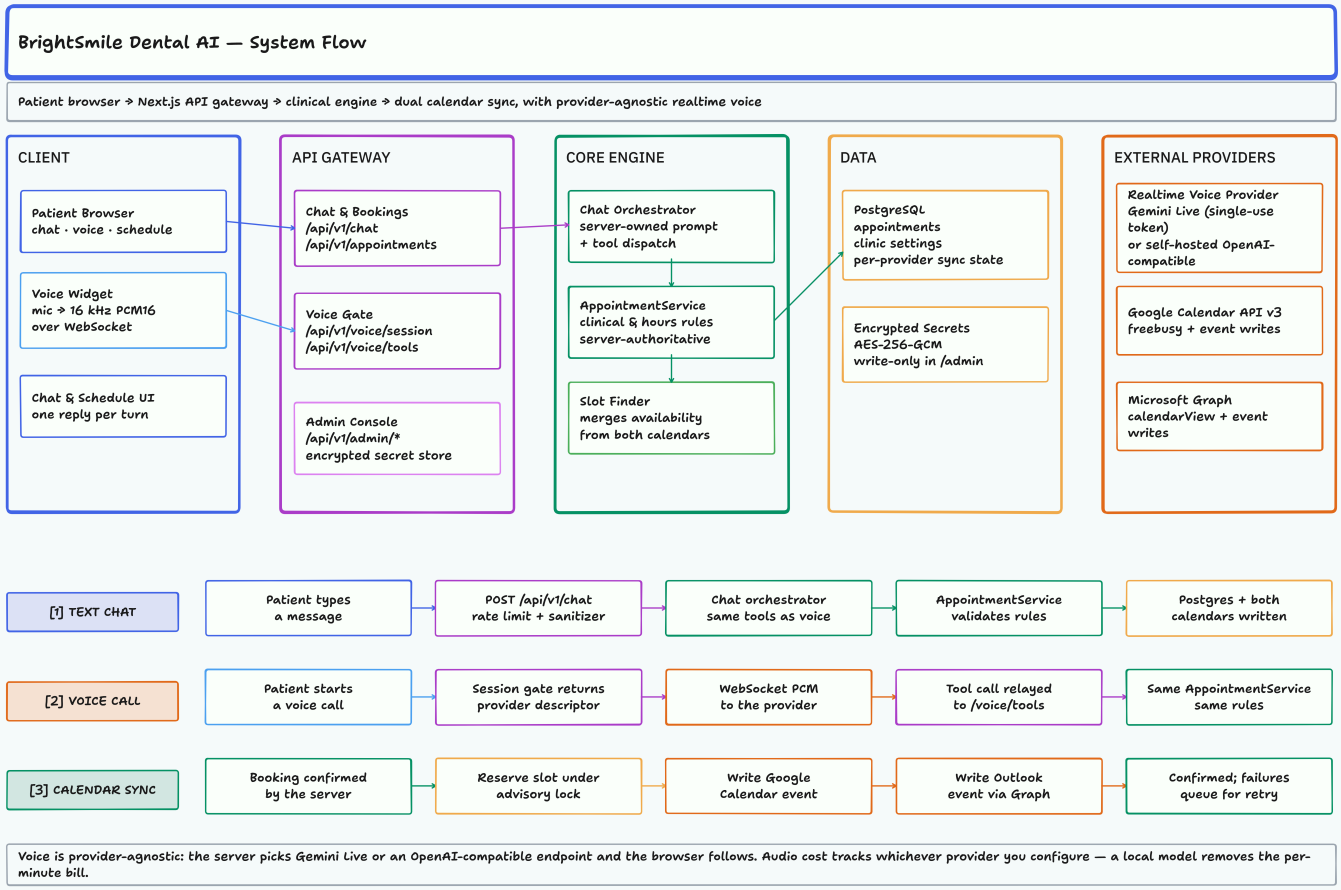
BrightSmile Dental AI Platform

Conversational voice and chat scheduling, clinical qualification gating, and dual-calendar synchronization — with the voice provider chosen by configuration, not hardcoded.

< 300 ms	16 / 24 kHz	3 & 5	2
TURN LATENCY TARGET (MODELED)	PCM INPUT / OUTPUT	DENTISTS & SERVICES	CALENDARS WRITTEN PER BOOKING

System Flow

ARCHITECTURE MAP & RUNTIME LANES



One browser, one Next.js gateway, one clinical engine. Text chat and voice reach the same `AppointmentService`; the server picks the voice provider and the browser follows. Every confirmed booking is written to both calendars.

Booking Lifecycle

INBOUND TO CALENDAR SETTLEMENT

STEP 01 • INBOUND
Patient Engagement
Patients speak over a WebSocket audio stream or type in the chat portal, in everyday language.

STEP 02 • PERIMETER
Edge Guardrails
Rate limiting and pre-LLM sanitizers screen prompt injection and medical-advice requests.

STEP 03 • GATEWAY
Session Broker
The server selects the voice provider and returns its session descriptor. With Gemini that is a single-use token; an OpenAI-compatible endpoint needs none.

STEP 04 • DOMAIN
Clinical Engine
Server-side rules match the request against practitioner qualification tiers, clinic hours, and duration.

STEP 05 • STORAGE
Atomic Commit
A PostgreSQL advisory lock per dentist and date prevents overlap, then both calendars are written with per-provider retry state.

CLINICAL SCOPE | VOICE & CALENDARS | COST DRIVERS

Scope, Sync & Operating Cost

Which dentist may perform which procedure, how a booking reaches both calendars, and what actually drives the monthly bill.

Practitioner Credential Tiers

HARD BOUNDARY, ENFORCED IN CODE

CODE	PROCEDURE	DURATION	ELIGIBLE PRACTITIONERS
A	**General Checkup & Cleaning**	60 min	All Junior, Senior 1, Senior 2
B	**Simple Tooth Extraction**	60 min	All Junior, Senior 1, Senior 2
C	**Root Canal Treatment**	150 min	Specialists Senior 1, Senior 2
D	**Dental Crown / Veneer**	120 min	Specialists Senior 1, Senior 2
E	**Full Mouth Implant Surgery**	360 min	Specialists 6 continuous hours

Deterministic Scope Enforcement

Qualification is enforced in `AppointmentService`, never by the model's prompt. If a patient asks for a root canal with Dr. Junior, the server raises `DentistQualificationError` and the assistant reroutes to an eligible specialist.

Voice & Calendar Mechanics

STREAMING & EVENTUAL CONVERGENCE

Realtime Voice

PCM16 audio streams over a WebSocket at 16 kHz up and 24 kHz down, with no separate speech-to-text hop. Tool calls are relayed to `/api/v1/voice/tools` for server-side validation. Master provider keys never reach the browser.

Dual Calendar

Google Calendar (API v3 FreeBusy) and Microsoft Graph (`calendarView`) sit behind one provider contract. A booking stays `pending` until both providers converge; per-provider retry state is drained by an authenticated reconcile endpoint.

Credential Handling

OAuth secrets live in an AES-256-GCM encrypted store managed from `/admin`, or in environment variables as a bootstrap fallback. The admin API returns configured status only, never the stored value.

What Drives the Monthly Bill

NO SINGLE TOTAL IS QUOTED, AND HERE IS WHY

COMPONENT	COST DRIVER
Realtime voice	Per audio-minute in and out on a hosted provider. On a local speech-to-speech model this line becomes electricity and GPU time.
Text chat LLM	Per input and output token on the configured model; scales with conversation length, not with bookings.
Application hosting	Serverless plan or container, sized to request volume.
PostgreSQL	Managed instance with automated backups and connection pooling.
Calendar APIs	Google Calendar and Microsoft Graph free quotas comfortably cover a single clinic.

Why voice deserves the scrutiny

Voice is the one line item large enough to distort a small clinic's software budget. Hosted realtime audio is priced per minute, so a practice handling a few thousand spoken minutes a month pays more for voice than for everything else combined.

That is a property of per-minute voice pricing, not of this application. The same code pointed at a self-hosted OpenAI Realtime-compatible server — for example **huggingface/speech-to-speech** — costs only electricity. Choose the provider you can actually fund; the application does not choose for you.

Managed Cloud vs. Self-Hosted Edge

COST & LATENCY, SIDE BY SIDE

DIMENSION	MANAGED CLOUD (HOSTED REALTIME)	SELF-HOSTED EDGE (LOCAL REALTIME SERVER)
Billing unit	Per audio-minute in and out, metered by the vendor.	Electricity for the GPU you already own. Nothing is metered per call.
Turn latency	Vendor inference plus a wide-area round trip. The `< 300 ms` figure above is a **modeled target**, not a measurement from this deployment.	STT, LLM and TTS on one local GPU with no vendor hop. The reference server logs per-response STT, LLM and first-audio TTS durations — read the number there rather than quoting one.
Capacity	Bounded by the provider's concurrency quota; raised by paying more.	Bounded by GPU memory. A single instance typically serves one concurrent session.
Credentials	The server mints a single-use ephemeral token per session.	Nothing to mint. The endpoint sits on your own network.
Failure mode	Provider outage or quota exhaustion, outside your control.	GPU out-of-memory or a crashed process that you restart.

Latency in both columns is a design target until measured on the hardware that serves patients. No figure here was produced by a benchmark run of this application.

VERIFICATION STATUS: All provider calls in the automated suite are mocked. Live voice and calendar behaviour requires separate validation with real credentials. — Validation required